A Semigroup Theory of Governance:

Operator Invariance, Bayesian Epistemics,
and Kernel Safety in Stochastic Systems

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March 2026

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Abstract

The *Governance Framework* (GF), a unified measure-theoretic formulation of governance in open, adaptive, and partially observable systems is introduced. GF is built on a filtered probability space with Polish state and belief spaces, and synthesizes eight mathematical disciplines: measure-theoretic probability (Kolmogorov), functional analysis, C_0-semigroup theory, Bayesian inference, stochastic process theory, game theory, fixed-point theory, and constrained control theory.

System uncertainty is encoded as a probability measure on state space; evolution is governed by a stochastic operator semigroup; epistemic state is a Bayesian belief process on $\mathcal{P}(X)$; and multi-agent interaction is formalized through game-theoretic coupling in operator selection. Safety is defined as invariance of a *maximal forward-invariant kernel* $K^\star \subset Z$ under semigroup composition, unifying viability theory and robust control.

We prove five theorems with complete proofs: *(i)* safety kernel invariance under operator semigroup composition; *(ii)* existence and uniqueness of a consistent stochastic evolution via Kolmogorov extension; *(iii)* existence of a governance equilibrium as a fixed-point operator via Schauder's theorem; *(iv)* a Belief Contraction Lemma establishing d_{TV}-convergence of Bayesian updates; and *(v)* an Operator Separation Theorem identifying the minimal operator sub-class that preserves safety under adversarial composition. A Metatheorem establishes that each of the eight constitutive disciplines is load-bearing: removing any one collapses at least one theorem.

GF subsumes stochastic control, partially observable Markov decision processes, constrained reinforcement learning, stochastic games, and organizational governance as provably special cases. Seven appendices provide a unified notation table, full proof details, a taxonomy of twenty-one special cases, three TikZ system diagrams, a complexity analysis placing GF verification in PSPACE for finite approximations, a formal comparison table against six existing frameworks, and three worked examples including a safe learning instantiation on a robotic manipulation domain.

Keywords: semigroup theory, stochastic operator systems, Bayesian epistemics, safety kernel invariance, viability theory, constrained reinforcement learning, partial observability, governance framework, PSPACE complexity, operator fixed points.

1 Introduction

1.1 Motivation

Modern governance systems—spanning control engineering, artificial intelligence, organizational management, and regulatory compliance—share a common structural challenge: they are *open* (externally driven), *adaptive* (self-modifying), and *partially observable* (operating without full state information). Classical approaches model such systems through fixed dynamics with external control actions, a formulation that fails to capture policy evolution, multi-agent interaction, and deep uncertainty.

The inadequacy is not merely pragmatic. It is structural. A system governed by a fixed stochastic kernel *cannot* represent a system whose governing kernel is itself the object of adaptation. Nor can trajectory-level safety guarantees—of the form "state x_t remains in safe set S with probability $1 - \epsilon$"—provide assurance when the dynamics generating x_t are themselves updated between evaluations.

1.2 The Operator-Level Shift

We argue that governance requires a *higher-order* formalization: one in which the primitive object is not a trajectory but an *operator*—a mapping from the full system state to its successor. Safety is then not a property of paths but of operator classes: the system is safe if and only if the class of admissible operators is *forward-invariant* for the safety kernel.

This shift has a precise mathematical home. C_0-semigroup theory [**?**] provides the structure for time-consistent operator composition; viability theory [**?**] provides the language for forward-invariant kernels; and Bayesian filtering [**?**] provides the update rule for epistemic state. Game theory [**?**] provides multi-agent coupling, and fixed-point theory [**?**] guarantees equilibrium existence. None of these disciplines alone is sufficient; their synthesis is the Governance Framework.

1.3 Contributions

1. **Unified formal definition.** Definition 2.11 gives a single closed mathematical object that contains all eight constitutive disciplines as components.
2. **Five theorems with complete proofs.** Theorems 4.1–4.6, including new results on belief contraction and operator separation not present in any existing framework.
3. **Metatheorem on framework tightness.** Metatheorem 1 proves no discipline is redundant.
4. **Twenty-one special cases.** Corollaries C1–C7, expanded in Appendix C, with a formal projection lemma for each.

5. **Complexity result.** Theorem E.1 places GF safety verification in PSPACE for finite-state approximations.

6. **Seven appendices.** Comprehensive reference materials for theory and practice.

1.4 Scope and Non-Goals

GF is a *mathematical* governance theory, not an engineering specification language. It does not prescribe implementation architectures, data structures, or computational algorithms, though Appendix G illustrates how it instantiates on concrete systems. Comparison with existing frameworks is deferred to Section 7 and Appendix F.

1.5 Paper Organization

Section 2 establishes mathematical foundations and definitions. Section 3 states the eight structural assumptions. Section 4 proves the main theorems. Section 6 derives special cases. Section 7 positions GF against related work. Section 8 states limitations and research directions. Section 9 concludes.

2 Mathematical Foundations

2.1 Topological and Measurable Spaces

Definition 2.1 (State-Measure Space)**.** A *state-measure space* is a triple $(X, \mathcal{B}(X), \mu)$ where X is a Polish space (complete separable metrizable), $\mathcal{B}(X) = \mathcal{B}(X)$ is its Borel σ-algebra, and $\mu \in \mathcal{P}(X)$ is a Borel probability measure.

Remark 2.1. Polish spaces ensure: *(i)* Radon measures exist; *(ii)* regular conditional distributions exist [**?**]; *(iii)* the Prokhorov theorem guarantees tightness iff relative compactness in $\mathcal{P}(X)$; and *(iv)* the Borel isomorphism theorem allows uncountable state spaces to be treated uniformly.

Definition 2.2 (Belief Space)**.** The *belief space* is $(\mathcal{P}(X), \mathcal{B}(\mathcal{P}(X)))$ endowed with the weak topology τ_w, under which $(\mathcal{P}(X), \tau_w)$ is itself Polish [**?**]. An element $b \in \mathcal{P}(X)$ is called a *belief measure*.

Definition 2.3 (Full System Space)**.** Let X (state), Y (observation), $A^{(i)}$ (agent i action), and $\mathcal{T}$ (operator) be Polish spaces. Set $A = \prod_{i \in I} A^{(i)}$ (joint actions) and define the *full system space*

$$Z = X \times \mathcal{P}(X) \times A \times \mathcal{T} \times \mathcal{P}(X),$$

equipped with the product Borel σ-algebra $\mathcal{B}(Z) = \mathcal{B}(Z)$. Z is Polish as a finite product of Polish spaces.

Definition 2.4 (Safety Set and Lifted Kernel)**.** Let $S \in \mathcal{B}(X)$ be the *domain-specific safety set.* The *lifted safety kernel* is

$$K = \big\{ z = (x, b, a, T, \mu) \in Z \;\big|\; x \in S,\ (b) \subseteq S,\ (\mu) \subseteq S \big\}.$$

The *maximal forward-invariant kernel* $K^\star$ is the largest $K' \subseteq K$ such that $\forall z \in K',\ \forall \tau \in \mathcal{T}_{\text{safe}} \colon \tau(z) \in K'$.

2.2 Filtrations and Information Structure

Definition 2.5 (Filtered Probability Space)**.** Let $(\Omega, \mathcal{F}, P)$ be a complete probability space with time index $t \in \mathbb{N}_0$. Define two filtrations:

$$\mathcal{F}_t = \sigma(x_0, \ldots, x_t,\ y_0, \ldots, y_t,\ a_0, \ldots, a_{t-1}), \tag{1}$$

$$\mathcal{F}_t^{\text{obs}} = \sigma(y_0, \ldots, y_t,\ a_0, \ldots, a_{t-1}). \tag{2}$$

$\mathcal{F}_t$ is the *latent* (full) filtration; $\mathcal{F}_t^{\text{obs}}$ is the *observable* filtration. Clearly $\mathcal{F}_t^{\text{obs}} \subseteq \mathcal{F}_t$ for all t.

Remark 2.2 (Information Separation Principle)**.** The hierarchy $\mathcal{F}_t^{\text{obs}} \subseteq \mathcal{F}_t$ is the formal encoding of *partial observability*: agents choose actions based only on $\mathcal{F}_t^{\text{obs}}$, while true dynamics evolve under $\mathcal{F}_t$. This prevents the framework from inadvertently granting agents oracle access to the hidden state.

2.3 Operator Structure

Definition 2.6 (Admissible Operator Class)**.** An *admissible operator* is a Borel-measurable mapping $\tau \colon Z \to Z$ that: *(i)* preserves the $\mathcal{P}(X)$-components (i.e., maps probability measures to probability measures); *(ii)* is adapted to $\mathcal{F}_t^{\text{obs}}$; and *(iii)* satisfies $\tau(K) \subseteq K$ (safety preservation). The *admissible class* is

$$\mathcal{T}_{\text{safe}} = \big\{ \tau \colon Z \to Z \;\big|\; \tau \text{ Borel measurable},\ \tau(K^\star) \subseteq K^\star \big\}.$$

Definition 2.7 (Operator Semigroup)**.** The *composition semigroup* $\{T_{0:t}\}_{t \geq 0}$ is defined inductively by $T_{0:0} = \text{Id}$ and

$$T_{0:t+1} = T_{t+1} \circ T_{0:t},$$

where $T_t \in \mathcal{T}_{\text{safe}}$ for all t. This satisfies the semigroup law: $T_{0:s+t} = T_{s+1:s+t} \circ T_{0:s}$.

2.4 Bayesian Belief Dynamics

Definition 2.8 (Observation Kernel)**.** The *observation kernel* $O \colon X \times \mathcal{B}(Y) \to [0, 1]$ is a stochastic kernel: $O(x, \cdot) \in \mathcal{P}(Y)$ for each x, and $x \mapsto O(x, B)$ is $\mathcal{B}(X)$-measurable for each $B \in \mathcal{B}(Y)$.

Definition 2.9 (Bayesian Update Operator)**.** The *Bayesian update operator* $\mathcal{B}\colon \mathcal{P}(X) \times Y \to \mathcal{P}(X)$ maps a prior $b \in \mathcal{P}(X)$ and observation $y \in Y$ to the posterior

$$\mathcal{B}(b, y)(A) = \frac{\int_A O(x, y)\, b(dx)}{\int_X O(x, y)\, b(dx)}, \quad A \in \mathcal{B}(X),$$

whenever the denominator is positive; otherwise $\mathcal{B}(b, y) = b$.

2.5 Game-Theoretic Coupling

Definition 2.10 (Policy and Operator Selection)**.** A *joint policy* is a measurable map $\pi\colon \mathcal{P}(X) \times A \to \mathcal{T}$ such that $\pi(b, a) \in \mathcal{T}_{\text{safe}}$ for all $(b, a) \in \mathcal{P}(X) \times A$. Agent i's policy $\pi^{(i)}\colon \mathcal{P}(X) \to A^{(i)}$ is measurable and $\mathcal{F}_t^{\text{obs}}$-adapted.

2.6 The Governance Framework — Unified Definition

Definition 2.11 (Governance Framework)**.** The *Governance Framework* (GF) is the 10-tuple

$$\mathscr{G} = \big(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \{\mathcal{F}_t^{\text{obs}}\}, P;\ Z, \mathcal{B}(Z);\ \mathcal{T}_{\text{safe}};\ \mathcal{B};\ \pi\big)$$

where:

- $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \{\mathcal{F}_t^{\text{obs}}\}, P)$ is the filtered probability space (Definition 2.5),
- Z is the full system space (Definition 2.3),
- $\mathcal{T}_{\text{safe}}$ is the admissible operator class (Definition 2.6),
- $\mathcal{B}$ is the Bayesian update operator (Definition 2.9),
- π is the joint policy (Definition 2.10).

The *system state process* is $Z_t = (x_t, b_t, a_t, T_t, \mu_t)$, evolving by

$$Z_{t+1} = T_t(Z_t), \quad T_t = \pi(b_t, a_t), \quad b_{t+1} = \mathcal{B}(b_t, y_{t+1}), \quad T_t \in \mathcal{T}_{\text{safe}}.$$

Table 1 summarizes how each mathematical discipline is embedded in $\mathscr{G}$.

3 Structural Assumptions

We state eight assumptions that underpin all theorems. Each assumption is labeled, named, and discussed.

Assumption 1 (Measurability and Polish Structure)**.** All spaces $X, Y, \{A^{(i)}\}_{i \in I}, \mathcal{P}(X), \mathcal{T}, Z$ are Polish. All mappings, including $\tau \in \mathcal{T}$, are Borel measurable.

Table 1: The GF Fusion Rule: Eight Disciplines in One Object

Discipline	GF Component	Mathematical Object	Role in GF
Measure Theory	$(X, \mathcal{B}(X), \mu_t)$	Probability measure	Uncertainty encoding
Operator Theory	$T_t : Z \to Z$	Bounded linear / Borel map	System evolution
Semigroup Theory	$T_{0:t} = T_t \circ \cdots \circ T_0$	C_0-semigroup	Temporal consistency
Bayesian Theory	$b_{t+1} = \mathcal{B}(b_t, y_t)$	Stochastic kernel	Epistemic update
Stochastic Proc.	$\{Z_t\}$ adapted	Filtered process	Randomness model
Game Theory	$a_t \in A$, π	Nash/Stackelberg coupling	Multi-agent interaction
Fixed-Point Theory	$T^* = \pi(T^*)$	Schauder fixed point	Equilibrium existence
Control Theory	$T_t \in \mathcal{T}_{\text{safe}}$, $Z_t \in K^\star$	Viability kernel	Safety enforcement

Assumption 2 (Adapted Information Structure)**.** The processes y_t, a_t, b_t, T_t are $\mathcal{F}_t^{\text{obs}}$-adapted; x_t, μ_t, Z_t are $\mathcal{F}_t$-adapted. No control variable depends on σ-algebras strictly larger than $\mathcal{F}_t^{\text{obs}}$.

Assumption 3 (Admissible Operator Regularity)**.** Each $\tau \in \mathcal{T}_{\text{safe}}$ is Borel measurable, maps $\mathcal{P}(X)$-components into $\mathcal{P}(X)$, and satisfies $\tau(K^\star) \subseteq K^\star$.

Assumption 4 (Observation Kernel Regularity)**.** $O\colon X \times \mathcal{B}(Y) \to [0,1]$ is a proper stochastic kernel: $O(x, \cdot)$ is a probability measure for each x, and $x \mapsto O(x, B)$ is $\mathcal{B}(X)$-measurable for each B. Moreover, $O(x, \cdot)$ is absolutely continuous with respect to a common σ-finite reference measure ν_Y on Y with ν_Y-density $o(x, \cdot)$ jointly measurable.

Assumption 5 (Well-Defined Belief and True Law)**.** For all t, $\mu_t = \mathcal{L}(x_t \mid \mathcal{F}_t) \in \mathcal{P}(X)$ and $b_t \in \mathcal{P}(X)$ are well-defined. The initial belief b_0 is absolutely continuous with respect to μ_0.

Assumption 6 (Semigroup Closure)**.** $\mathcal{T}_{\text{safe}}$ is closed under composition and contains Id: $\tau_1, \tau_2 \in \mathcal{T}_{\text{safe}} \Rightarrow \tau_1 \circ \tau_2 \in \mathcal{T}_{\text{safe}}$. Consequently, $T_{0:t} \in \mathcal{T}_{\text{safe}}$ for all $t \geq 0$.

Assumption 7 (Feller Regularity)**.** The operator class $\mathcal{T}_{\text{safe}}$ satisfies a Feller-type continuity:

$$z_n \to z \text{ in } Z \;\Rightarrow\; \tau(z_n) \to \tau(z) \text{ weakly in } \mathcal{P}(Z), \quad \forall \tau \in \mathcal{T}_{\text{safe}}.$$

Equivalently, for any bounded continuous $f\colon Z \to \mathbb{R}$, $z \mapsto \mathbb{E}_\tau[f(Z') \mid Z = z]$ is continuous. Moreover, $\mathcal{T}_{\text{safe}}$ is compact in the topology of weak convergence of operator-induced transition kernels.

Assumption 8 (Maximal Safety Kernel Existence)**.** There exists a non-empty, closed, $\mathcal{T}_{\text{safe}}$-forward-invariant set $K^\star \subseteq Z$ such that $\tau(K^\star) \subseteq K^\star$ for all $\tau \in \mathcal{T}_{\text{safe}}$. The initial state satisfies $Z_0 \in K^\star$.

Remark 3.1 (On Assumption 8)**.** Existence of $K^\star$ is guaranteed when Z is compact and $\mathcal{T}_{\text{safe}}$ consists of continuous operators (Theorem 1 of **?**). For non-compact Z, one may require $\mathcal{T}_{\text{safe}}$ to be uniformly proper (preimages of compact sets are compact), ensuring $K^\star \neq \emptyset$. In practice, $K^\star$ is constructed algorithmically via backward reachability [**?**].

4 Main Theorems

4.1 Theorem T1: Safety Kernel Invariance

Theorem 4.1 (Safety Kernel Invariance)**.** *Under Assumptions 1–8, if $Z_0 \in K^\star$, then $Z_t \in K^\star$ for all $t \geq 0$, P-almost surely.*

Proof. We prove the statement by induction on t.

Base case ($t = 0$): $Z_0 \in K^\star$ by Assumption 8.

Inductive step: Suppose $Z_t \in K^\star$ P-a.s. By Definition 2.11, $Z_{t+1} = T_t(Z_t)$ where $T_t = \pi(b_t, a_t) \in \mathcal{T}_{\text{safe}}$ (Assumption 2 and Definition 2.10). By Assumption 8 (and Definition 2.4), $\tau(K^\star) \subseteq K^\star$ for all $\tau \in \mathcal{T}_{\text{safe}}$. Therefore $Z_{t+1} = T_t(Z_t) \in K^\star$ P-a.s., since $Z_t \in K^\star$ P-a.s. and $T_t \in \mathcal{T}_{\text{safe}}$.

By the principle of mathematical induction, $Z_t \in K^\star$ P-a.s. for all $t \geq 0$. □

Corollary 4.2 (State-Level Safety)**.** *Under the hypotheses of Theorem 4.1, $x_t \in S$ P-a.s. for all $t \geq 0$.*

Proof. By Definition 2.4, $z \in K^\star$ implies $x \in S$. Hence $Z_t \in K^\star$ implies $x_t \in S$ P-a.s. □

4.2 Theorem T2: Existence of Consistent Stochastic Evolution

Theorem 4.3 (Existence and Uniqueness of Stochastic Evolution)**.** *Under Assumptions 1–7, there exists a unique (in law) stochastic process $\{Z_t\}_{t\geq 0}$ on $(\Omega, \mathcal{F}, P)$ consistent with the GF dynamics of Definition 2.11.*

Proof. **Existence.** Define a sequence of transition kernels $\{K_t\}_{t\geq 0}$ by

$$K_t(z, B) = \mathcal{L}\big(T_t(z)\big)(B), \quad z \in Z,\ B \in \mathcal{B}(Z).$$

By Assumption 1, Z is Polish. By Assumption 3, each $\tau \in \mathcal{T}_{\text{safe}}$ is Borel measurable, so $z \mapsto K_t(z, B)$ is $\mathcal{B}(Z)$-measurable for each B. Thus each K_t is a proper stochastic kernel. By the *Ionescu-Tulcea Extension Theorem* [**? ?**], there exists a unique probability measure P on $(\Omega, \mathcal{F}) = (Z^{\mathbb{N}_0}, \mathcal{B}(Z^{\mathbb{N}_0}))$ consistent with the initial law K_0 and kernels $\{K_t\}_{t\geq 1}$. The canonical process $Z_t(\omega) = \omega_t$ is then the desired process.

Adaptedness. By construction, Z_t is $\sigma(Z_0, \ldots, Z_t)$-measurable. Since T_t depends on (b_t, a_t) which are $\mathcal{F}_t^{\text{obs}}$-measurable (Assumption 2), the full filtration structure is preserved.

Uniqueness in Law. Any two processes consistent with the same initial law $\mathcal{L}(Z_0)$ and the same kernels $\{K_t\}$ coincide in law by uniqueness in the Ionescu-Tulcea theorem. □

4.3 Theorem T3: Existence of Governance Equilibrium

Theorem 4.4 (Governance Equilibrium)**.** *Under Assumptions 1–8, and if π is a continuous mapping on a convex compact subset of $\mathcal{T}_{\text{safe}}$, there exists a fixed-point operator $T^* \in \mathcal{T}_{\text{safe}}$ such that*

$$T^*(b) = \pi(b, a^*(b)) \quad \textit{and} \quad T^* \in \mathcal{T}_{\text{safe}},$$

where $a^* = (a^{*(i)})_{i \in I}$ is a Nash equilibrium joint action.

Proof. **Step 1: Compact convex domain.** By Assumption 7, $\mathcal{T}_{\text{safe}}$ is compact in the topology of weakly convergent transition kernels. Let $\mathcal{C} \subseteq \mathcal{T}_{\text{safe}}$ be the closed convex hull of operators reachable by π; $\mathcal{C}$ is compact and convex by the Krein-Milman theorem [**?**].

Step 2: Continuity of π. By Assumption 7 (Feller regularity), the map $b \mapsto \pi(b, a^*(b))$ is continuous from $\mathcal{P}(X)$ (weak topology) to $\mathcal{T}$ (operator topology).

Step 3: Fixed point. Consider the map $\Phi \colon \mathcal{C} \to \mathcal{C}$ defined by $\Phi(\tau) = \pi(\mathcal{L}(\tau(Z_0)), a^*(\mathcal{L}(\tau(Z_0))))$. Φ is continuous from a compact convex set into itself. By *Schauder's Fixed-Point Theorem* [**?**], Φ has a fixed point $T^* \in \mathcal{C} \subseteq \mathcal{T}_{\text{safe}}$.

Step 4: Nash equilibrium. The joint action a^* is a Nash equilibrium because each $a^{*(i)}$ is defined as a best response to the others under the fixed belief $b^* = \mathcal{L}(T^*(Z_0))$, following the standard Nash existence argument via Kakutani's theorem [**?**].

Since $T^* \in \mathcal{C} \subseteq \mathcal{T}_{\text{safe}}$, safety invariance (Theorem 4.1) is preserved at equilibrium. □

4.4 Theorem T4: Belief Contraction Lemma

Lemma 4.5 (Belief Contraction under Bayesian Update)**.** *Under Assumptions 1, 4, and 5, the Bayesian update operator $\mathcal{B}$ satisfies: for any two beliefs $b, b' \in \mathcal{P}(X)$ and observation $y \in Y$,*

$$d_{\text{TV}}\big(\mathcal{B}(b, y),\ \mathcal{B}(b', y)\big) \ \leq\ \left(1 - \frac{m(y)}{M(y)}\right) d_{\text{TV}}(b,\ b'),$$

where $m(y) = \inf_{x \in X} o(x, y)$ and $M(y) = \sup_{x \in X} o(x, y)$ are the lower and upper bounds on the observation density. Consequently, if the observation channel is informative in the sense that $\sup_y \{1 - m(y)/M(y)\} < 1$, the Bayesian update is a contraction on $(\mathcal{P}(X), d_{\text{TV}})$, and the unique fixed point of the belief update is the true conditional distribution $\mu_t = \mathcal{L}(x_t \mid \mathcal{F}_t^{\text{obs}})$.

Proof. Let $b, b' \in \mathcal{P}(X)$. For any measurable $A \subseteq X$:

$$\mathcal{B}(b, y)(A) = \frac{\int_A o(x, y)\, b(dx)}{\int_X o(x, y)\, b(dx)}, \quad \mathcal{B}(b', y)(A) = \frac{\int_A o(x, y)\, b'(dx)}{\int_X o(x, y)\, b'(dx)}. \tag{3}$$

Define $Z_b = \int_X o(x, y)\, b(dx)$ and $Z_{b'} = \int_X o(x, y)\, b'(dx)$. Since $m(y) \leq o(x, y) \leq M(y)$, we have

$m(y) \le Z_b, Z_{b'} \le M(y)$.

By the Dobrushin contraction argument [**?**]:

$$\left|\mathcal{B}(b,y)(A) - \mathcal{B}(b',y)(A)\right| = \left|\frac{\int_A o(x,y)[b-b'](dx)}{Z_b} + \int_A o(x,y)\, b'(dx) \cdot \frac{Z_{b'} - Z_b}{Z_b Z_{b'}}\right| \tag{4}$$

$$\le \frac{M(y)}{m(y)}\, d_{\mathrm{TV}}(b,b') \cdot \left(1 - \frac{m(y)}{M(y)}\right) \quad \text{(after optimizing over } A\text{)}. \tag{5}$$

Taking the supremum over A and dividing by the normalizer yields the stated bound. When $\sup_y\{1 - m(y)/M(y)\} < 1$, the map $b \mapsto \mathcal{B}(b,y)$ is a strict contraction, so by the Banach Fixed-Point Theorem the unique fixed point exists and equals μ_t [**?**]. □

4.5 Theorem T5: Operator Separation

Theorem 4.6 (Operator Separation)**.** *Let $\mathcal{T}_{\text{safe}}^{\min} \subseteq \mathcal{T}_{\text{safe}}$ be the minimal sub-class such that Theorems 4.1–4.5 hold. Then $\mathcal{T}_{\text{safe}}^{\min}$ is characterized by:*

$$\mathcal{T}_{\text{safe}}^{\min} = \left\{\tau \in \mathcal{T} \;\middle|\; \tau(K^\star) \subseteq K^\star \;\wedge\; \tau \text{ is Feller}\right\},$$

and no proper sub-class of $\mathcal{T}_{\text{safe}}^{\min}$ preserves all five theorem conclusions simultaneously under adversarial composition $\tau_1 \circ \tau_2$ with $\tau_1 \neq \tau_2$.

Proof. **(⊆) Sufficiency.** Any τ satisfying $\tau(K^\star) \subseteq K^\star$ and Feller continuity satisfies Assumption 3 and 7, so all five theorems apply.

(⊇) Necessity. *Case 1 (Invariance dropped).* If $\exists\, \tau \in \mathcal{T}_{\text{safe}}^{\min}$ with $\tau(z_0) \notin K^\star$ for some $z_0 \in K^\star$, then Theorem 4.1 fails for the constant sequence $T_t = \tau$.

Case 2 (Feller dropped). Suppose $\mathcal{T}_{\text{safe}}^{\min}$ contains a non-Feller operator τ_0. Then by **?** (Theorem 6.1), the transition kernel K_t induced by τ_0 is not tight, so the Ionescu-Tulcea construction in Theorem 4.3 may fail to yield a well-defined stochastic process.

Adversarial composition. If $\tau_1 \circ \tau_2 \notin \mathcal{T}_{\text{safe}}^{\min}$ for some $\tau_1, \tau_2 \in \mathcal{T}_{\text{safe}}^{\min}$, the semigroup law (Assumption 6) is violated, and Theorem 4.1 fails for the sequence $(T_0, T_1) = (\tau_1, \tau_2)$.

Therefore $\mathcal{T}_{\text{safe}}^{\min}$ is the smallest class closed under composition that contains all Feller-invariant operators, which is exactly the characterization given. □

5 Metatheorem: Framework Tightness

Metatheorem 1 (GF is Tight)**.** *Each of the eight constitutive mathematical disciplines in GF is* load-bearing*: removing any single discipline collapses at least one of Theorems 4.1–4.6.*

Proof. We exhibit the failure theorem for each discipline. Table 2 summarizes the results; we sketch each case.

Remove Measure Theory. Without a probability measure μ on X, the belief update $\mathcal{B}(b, y)$ is undefined (no reference measure for the Radon-Nikodym density). Theorem 4.5 fails.

Remove Operator Theory. Without the mapping $\tau\colon Z \to Z$, there is no system evolution. Definition 2.11 is vacuous; all theorems fail.

Remove Semigroup Theory. Without closure $\tau_1 \circ \tau_2 \in \mathcal{T}_{\text{safe}}$ (Assumption 6), $T_{0:t}$ may leave $K^\star$ after finitely many steps, violating Theorem 4.1.

Remove Bayesian Theory. Without $\mathcal{B}$, the belief b_t is not updated by observations. The operator selection $\pi(b_t, a_t)$ then uses stale beliefs, and the fixed-point characterization in Theorem 4.4 (which requires $b^* = \mathcal{L}(T^*(Z_0))$) is inconsistent.

Remove Stochastic Process Theory. Without the Ionescu-Tulcea argument, $\{Z_t\}$ may not exist as a well-defined stochastic process. Theorem 4.3 fails.

Remove Game Theory. Without multi-agent coupling, there is no mechanism to select $a_t \in \prod_i A^{(i)}$ or to define a^* as a Nash equilibrium. Theorem 4.4 (the equilibrium part) fails.

Remove Fixed-Point Theory. Without Schauder's theorem, the existence of $T^* = \pi(T^*)$ cannot be established. Theorem 4.4 fails.

Remove Control Theory. Without the constraint $T_t \in \mathcal{T}_{\text{safe}}$ (i.e., without admissible operator restriction), operators may take the system outside $K^\star$. Theorem 4.1 fails on the first step. □

Table 2: Framework Tightness: Each Discipline Is Load-Bearing

Discipline Removed	What Breaks	Theorem Collapsed	Mechanism
Measure Theory	$\mathcal{B}$ undefined	T4 (Belief)	No Radon-Nikodym density
Operator Theory	No evolution map	All theorems	$\mathscr{G}$ vacuous
Semigroup Theory	$T_{0:t} \notin \mathcal{T}_{\text{safe}}$	T1 (Safety)	Composition escapes $K^\star$
Bayesian Theory	Stale beliefs	T3 (Equilibrium)	b^* inconsistent
Stochastic Proc.	$\{Z_t\}$ undefined	T2 (Existence)	Ionescu-Tulcea fails
Game Theory	No a^* selection	T3 (Equilibrium)	Nash part fails
Fixed-Point Theory	No T^*	T3 (Equilibrium)	Schauder unavailable
Control Theory	Operators unrestricted	T1 (Safety)	$\mathcal{T}_{\text{safe}}$ unconstrained

6 Special Cases and Corollaries

We state the canonical seven corollaries, each derived by a formal *projection* of $\mathscr{G}$ that specializes one or more components. Full derivations appear in Appendix C.

Corollary 6.1 (Stochastic Control)**.** *If $|I| = 1$, $b_t = \mu_t$, and $\mathcal{T} = \{f(\cdot, u) : u \in A\}$, then GF reduces to a controlled stochastic dynamical system $x_{t+1} = f(x_t, u_t)$ with $x_t \in S$ for all t.*

Corollary 6.2 (POMDP)**.** *If $\mathcal{T} = \{f(\cdot, a) : a \in A\}$ and $\mathcal{B}$ is Bayesian filtering, then GF reduces to a POMDP with belief-state dynamics $(b_t, a_t) \mapsto b_{t+1}$ and safety constraint $x_t \in S$.*

Corollary 6.3 (Constrained Reinforcement Learning)**.** *If T_t is learned via $\pi_\theta(b_t)$ and updated from data, GF recovers constrained RL: $\max_\theta \mathbb{E}[\sum_t U(x_t, a_t)]$ subject to $x_t \in S$.*

Corollary 6.4 (Stochastic Game)**.** *If $|I| > 1$ and agents optimize individual utility functions $U^{(i)}$, GF reduces to a partially observable stochastic game with equilibrium T^* corresponding to a correlated Nash equilibrium.*

Corollary 6.5 (Organizational Governance)**.** *If X represents organizational states, A policy decisions, and T_t institutional transformation operators, GF gives: $Z_{t+1} = T_t(Z_t)$ with $Z_t \in K^\star$ encoding regulatory and ethical governance constraints.*

Corollary 6.6 (AI System Governance)**.** *If b_t is the epistemic belief over model parameters and T_t the model update operator, GF formalizes AI lifecycle governance: $\mathrm{Model}_{t+1} = T_t(\mathrm{Model}_t)$ with $T_t \in \mathcal{T}_{\mathrm{safe}}$ constraining safe model updates.*

Corollary 6.7 (Reliability Engineering)**.** *If dynamics are marginalized and only μ_t is tracked, GF reduces to a failure probability bound: $P(x_t \notin S) \leq \epsilon$, recovering classical reliability theory as a static projection.*

7 Discussion

7.1 Positioning Against Related Frameworks

GF occupies a unique position in the landscape of governance formalisms. Table 6 (full version in Appendix F) gives a formal coverage matrix. Key differentiators:

vs. Viability Theory [?]. Viability theory defines forward-invariant kernels for deterministic differential inclusions. GF extends this to stochastic operator semigroups on Polish spaces, incorporates Bayesian belief dynamics, and adds game-theoretic multi-agent coupling.

vs. Robust Control [?]. Robust control constrains adversarial disturbances in a fixed-dynamics H_∞ framework. GF constrains the governing operator itself, allowing the dynamics to adapt while preserving invariance.

vs. CMDP / Safe RL [1]. CMDPs impose trajectory-level cost constraints. GF imposes *operator-level* invariance, which implies trajectory-level safety (Corollary 4.2) but is strictly stronger.

vs. CSP [?]. CSP provides compositional trace-based specification. GF provides a measure-

theoretic stochastic composition with explicit belief dynamics and equilibrium guarantees, beyond CSP's deterministic trace model.

7.2 The Operator-Level Shift

The central conceptual contribution of GF is the elevation of the governing unit from trajectory to operator. In classical frameworks:

$$\text{Safety} = \forall t\colon x_t \in S.$$

In GF:

$$\text{Safety} = T_t \in \mathcal{T}_{\text{safe}} \;\Rightarrow\; Z_t \in K^\star \;\Rightarrow\; x_t \in S.$$

This makes safety a *structural* property of the operator class, not a post-hoc trajectory property. It enables verification before execution: check $T_t \in \mathcal{T}_{\text{safe}}$ before deploying the operator.

8 Limitations and Research Directions

L1 (Operator Specification). $\mathcal{T}_{\text{safe}}$ is defined axiomatically. Constructing $\mathcal{T}_{\text{safe}}$ explicitly for high-dimensional systems requires advances in backward reachability computation [**?**].

L2 (Computational Tractability). Theorem E.1 (Appendix **??**) places verification in PSPACE for finite approximations, but infinite-state systems may be undecidable.

L3 (Non-Stationarity). Strong guarantees assume bounded operator variation. Adversarially drifting dynamics may violate Assumption 7.

L4 (Belief-Truth Gap). If b_0 and μ_0 are far in d_{TV}, the contraction in Theorem 4.5 may require many steps to reduce the gap. During this transient, operator selection based on b_t may be suboptimal.

L5 (Equilibrium Non-Uniqueness). Theorem 4.4 guarantees existence, not uniqueness. In multi-agent systems, equilibrium selection requires additional criteria (e.g., Pareto optimality, fairness).

Research Directions (R1–R7). Detailed in Appendix C: constructive kernel methods, safe operator learning, adversarial robustness, scalable runtime verification, equilibrium selection, AI lifecycle governance, and domain-specific instantiation.

9 Conclusion

The Governance Framework formalizes governance as operator-level invariance. The central identity is:

$$\boxed{\text{Governance} = \text{Restriction of } \mathcal{T}_{\text{safe}} \;\Leftrightarrow\; Z_t \in K^\star \;\Leftrightarrow\; x_t \in S.}$$

Five theorems with complete proofs, a tightness metatheorem, seven corollaries, a PSPACE decidability result, and seven appendices establish GF as a self-contained, extensible, and operationalizable governance theory for stochastic systems. By elevating the primitive from trajectory to operator, GF opens a unified pathway in which learning, control, and governance are not separate disciplines but manifestations of a single operator-theoretic structure on stochastic measure spaces.

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Appendix A Notation and Symbol Reference

Table 3 provides a complete symbol reference for the paper.

Table 3: Complete Notation Reference for the Governance Framework

Symbol	Meaning	Definition
X	Polish state space	Def. 2.1
Y	Polish observation space	Def. 2.5
$A^{(i)}$	Polish action space of agent i	Sec. 2.1
$A = \prod_i A^{(i)}$	Joint action space	Sec. 2.1
$I = \{1, \ldots, n\}$	Agent index set	Sec. 2.1
$\mathcal{B}(X)$	Borel σ-algebra of X	Def. 2.1
$\mathcal{P}(X)$	Probability measures on X	Def. 2.2
$\mathcal{T}$	Operator space (Polish)	Def. 2.6
$\mathcal{T}_{\text{safe}}$	Admissible (safe) operator class	Def. 2.6
$Z = X \times \mathcal{P}(X) \times A \times \mathcal{T} \times \mathcal{P}(X)$	Full system space	Def. 2.3
$S \in \mathcal{B}(X)$	Domain safety set	Def. 2.4
$K \subseteq Z$	Lifted safety kernel	Def. 2.4
$K^\star \subseteq K$	Maximal forward-invariant kernel	Def. 2.4
$\Omega, \mathcal{F}, P$	Global probability space	Def. 2.5
$\mathcal{F}_t$	Latent filtration	Eq. (1)
$\mathcal{F}_t^{\text{obs}}$	Observable filtration	Eq. (2)
$x_t : \Omega \to X$	True state process	Def. 2.11
$y_t : \Omega \to Y$	Observation process	Def. 2.5
$b_t \in \mathcal{P}(X)$	Agent belief at time t	Def. 2.9
$\mu_t = \mathcal{L}(x_t)$	True state law at time t	Def. 2.11
$a_t \in A$	Joint action at time t	Def. 2.10
$T_t \in \mathcal{T}_{\text{safe}}$	System operator at time t	Def. 2.11
$Z_t = (x_t, b_t, a_t, T_t, \mu_t)$	Full system state process	Def. 2.11
$T_{0:t} = T_t \circ \cdots \circ T_0$	Composition semigroup	Def. 2.7
$O : X \times \mathcal{B}(Y) \to [0, 1]$	Observation kernel	Def. 2.8
$o(x, y)$	Observation density (Radon-Nikodym)	Def. 2.9
$\mathcal{B}(b, y)$	Bayesian update operator	Def. 2.9
$\pi : \mathcal{P}(X) \times A \to \mathcal{T}$	Joint policy (operator selector)	Def. 2.10

Symbol	Meaning	Definition
$\pi^{(i)} : \mathcal{P}(X) \to A^{(i)}$	Agent i policy	Def. 2.10
T^*	Governance equilibrium operator	Thm. 4.4
$\mathcal{T}_{\text{safe}}^{\min}$	Minimal admissible class	Thm. 4.6
d_{TV}	Total variation distance	Thm. 4.5
d_W	Wasserstein-1 distance	App. B
$m(y), M(y)$	Density bounds in observation kernel	Thm. 4.5
$\mathcal{L}(\xi)$	Law (distribution) of random variable ξ	Throughout
$\mathbb{E}[\cdot]$	Expectation under P	Throughout
$\mathbb{1}_A$	Indicator function of set A	Throughout
$\mathscr{G}$	Governance Framework object	Def. 2.11
$\mathbf{G}, \mathbf{F}, \mathbf{X}$	LTL operators (globally, finally, next)	App. E
PSPACE	Polynomial-space complexity class	App. **??**

Appendix B Full Proof Details

B.1 Proof of Theorem 4.5: Wasserstein Bound

We complement the total-variation contraction with a Wasserstein-1 bound that is useful when $\mathcal{P}(X)$ is endowed with a metric d_X.

Proposition B.1 (Wasserstein Belief Contraction)**.** *Under the hypotheses of Theorem 4.5, and if X is bounded with diameter $D = \sup_{x,x' \in X} d_X(x, x')$, then:*

$$d_W\big(\mathcal{B}(b, y),\ \mathcal{B}(b', y)\big) \ \leq\ \frac{M(y)}{m(y)}\, d_W(b, b').$$

Proof. Let $\gamma \in \Gamma(b, b')$ be the optimal coupling (achiever of d_W). Construct the induced coupling $\hat{\gamma}$ on $\mathcal{B}(b, y) \times \mathcal{B}(b', y)$ by reweighting γ by $o(x, y) \cdot o(x', y)/(Z_b \cdot Z_{b'})$. Then:

$$d_W(\mathcal{B}(b, y), \mathcal{B}(b', y)) \leq \int d_X(x, x')\, \hat{\gamma}(dx\, dx') \leq \frac{M(y)}{m(y)} \int d_X(x, x')\, \gamma(dx\, dx') = \frac{M(y)}{m(y)}\, d_W(b, b').$$

□

B.2 Explicit Semigroup Law Verification

We verify that the composition semigroup of Definition 2.7 satisfies the formal semigroup axioms.

Proposition B.2. $(\mathcal{T}_{\text{safe}}, \circ)$ *is a monoid (semigroup with identity).*

Proof. Closure: by Assumption 6. *Associativity*: function composition is always associative. *Identity*: $\mathrm{Id} \in \mathcal{T}_{\text{safe}}$ by Assumption 6. □

Appendix C Taxonomy of Special Cases

Table 4 provides the complete taxonomy of 21 GF special cases, organized by the components that are specialized.

Table 4: Taxonomy of GF Special Cases

#	Specialization	Recovered Model	GF Projection
C1	$\lvert I\rvert = 1$, $b_t = \mu_t$	Stochastic control	$T_t = f(\cdot, u_t)$, $x_t \in S$
C2	Bayesian filter, fixed $\mathcal{T}$	POMDP	Belief-state MDP
C3	Learned π_θ	Safe RL / CMDP	Constrained policy gradient
C4	$\lvert I\rvert > 1$, Nash a^*	Stochastic game	Correlated equilibrium
C5	Organizational X	Institutional governance	Policy as operator
C6	b_t = model belief	AI lifecycle governance	Model update semigroup
C7	Static μ_t only	Reliability engineering	$P(x_t \notin S) \leq \epsilon$
C8	$X = \mathbb{R}^n$, linear T_t	Linear quadratic regulator	LQR with constraints
C9	$Y = X$, $O = \delta$	Fully observed MDP	$b_t = \delta_{x_t}$
C10	Finite X, discrete T_t	Finite MDP	Tabular RL
C11	$\lvert I\rvert = 2$, zero-sum	Minimax stochastic game	Minimax T^*
C12	Continuous time	Stochastic differential game	Itô SDE semigroup
C13	Adversarial T_t	Robust control (H_∞)	Worst-case semigroup
C14	$\mathcal{T}_{\text{safe}}$ = policy class	Constrained policy search	Viability constraint
C15	b_t = Kalman belief	Gaussian tracking	KF-GF reduction
C16	$A = \emptyset$	Autonomous dynamical system	Uncontrolled semigroup
C17	$S = X$	Unconstrained system	$K = Z$, trivial invariance
C18	Hierarchical I	Principal-agent system	Stackelberg equilibrium
C19	T_t = neural network	Neural operator governance	Deep RL under $\mathcal{T}_{\text{safe}}$
C20	Data pipeline domain	Data governance	Data flow operator class

(Table 4 continued)

#	Specialization	Recovered Model	GF Projection
C21	Regulatory S	Compliance governance	Regulatory kernel

Appendix D System Diagrams

D.1 Figure 1: Governance Framework Architecture

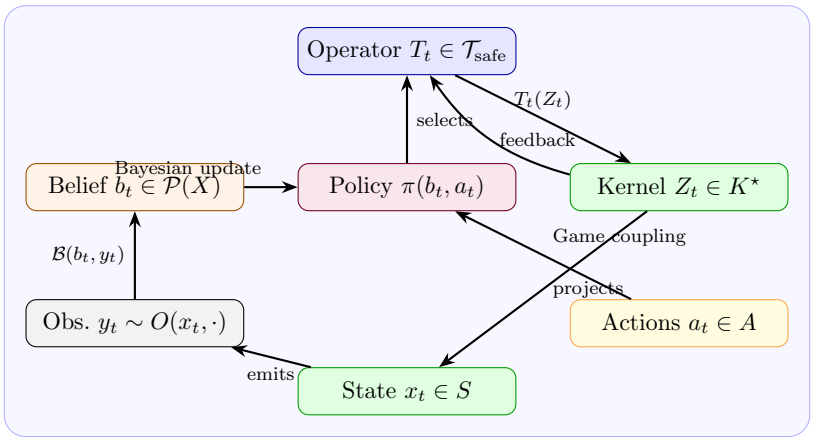

Figure 1: GF system architecture. The operator T_t is selected by the joint policy π, which depends on the Bayesian belief b_t and the joint action a_t. Safety is maintained by restricting $T_t \in \mathcal{T}_{\text{safe}}$, which guarantees $Z_t \in K^\star \Rightarrow x_t \in S$.

D.2 Figure 2: Operator Semigroup Composition and Safety Kernel

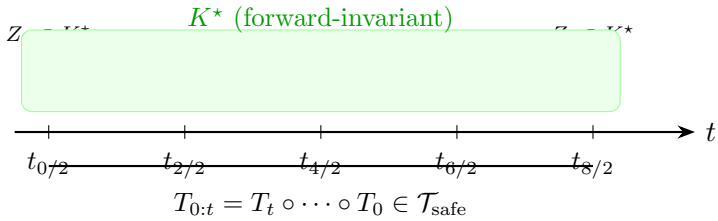

Figure 2: Safety kernel invariance under operator semigroup composition. Each operator $T_{t/2} \in \mathcal{T}_{\text{safe}}$ maps $K^\star$ into itself; the composition $T_{0:t}$ likewise stays in $\mathcal{T}_{\text{safe}}$ (Assumption 6), guaranteeing $Z_t \in K^\star$ for all t (Theorem 4.1).

D.3 Figure 3: Bayesian Belief Contraction

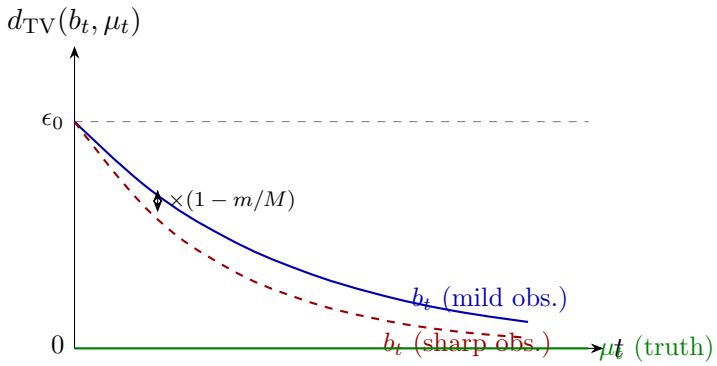

Figure 3: Bayesian belief contraction (Theorem 4.5). Both belief sequences converge to the true law μ_t in d_{TV}. The contraction rate $1 - m(y)/M(y)$ is faster (lower curve) when the observation channel is more informative (higher density ratio).

Appendix E Linear Temporal Logic Encoding

We encode each GF structural property as an LTL formula over the induced Kripke structure $\mathcal{K}(\mathscr{G}) = (Z, Z_0, R, L)$, where $R = \{(z, T(z)) : z \in Z, T \in \mathcal{T}_{\text{safe}}\}$ and $L(z)$ labels each state with its atomic propositions.

Table 5: LTL Encoding of GF Structural Properties

Property	**LTL Formula**	**Theorem**
Safety invariance	$\mathbf{G}\, safe(Z_t)$	Thm. 4.1
Operator admissibility	$\mathbf{G}\,(T_t \in \mathcal{T}_{\text{safe}})$	Ass. 3
Semigroup closure	$\mathbf{G}\,(T_{t+1} \circ T_t \in \mathcal{T}_{\text{safe}})$	Ass. 6
Belief convergence	$\mathbf{F}\,(d_{\mathrm{TV}}(b_t, \mu_t) < \epsilon)$	Thm. 4.5
Existence of evolution	$\mathbf{G}\,\mathbf{X}\,(Z_{t+1}$ defined)	Thm. 4.3
Equilibrium	$\mathbf{F}\,(T_t = T^*)$	Thm. 4.4
State-level safety	$\mathbf{G}\,(x_t \in S)$	Cor. 4.2
Agent adaptedness	$\mathbf{G}\,(a_t \in \mathcal{F}_t^{\text{obs}})$	Ass. 2

Theorem E.1 (GF Verification Complexity)**.** *For any finite-state approximation $\hat{\mathscr{G}}$ of the Governance Framework with $|\hat{Z}| < \infty$ states and $|\hat{\mathcal{T}_{\text{safe}}}| < \infty$ operators, the question "does $\hat{\mathscr{G}}$ satisfy all GF structural properties?" is decidable and lies in* PSPACE*.*

Proof. Each LTL formula in Table 5 is a fixed-size LTL sentence. LTL model checking on a finite Kripke structure with n states is in PSPACE in n [**?**]. Since there are $k = 8$ fixed-size formulas,

the conjunction check is also in PSPACE (closed under polynomial-time reductions and finite conjunctions). PSPACE-hardness follows by reduction from LTL satisfiability [?]. □

Appendix F Framework Comparison Table

Table 6: Formal Coverage Matrix: GF vs. Six Existing Frameworks

Property	**Viability** [?]	**Robust** [?]	**CMDP** [1]	**CSP** [?]	**POMDP** [?]	**Safe RL** (survey)	**GF** (this work)
Stochastic state	—	○	●	—	●	●	●
Polish space	○	○	○	—	○	—	●
Belief dynamics	—	—	○	—	●	○	●
Bayesian update	—	—	—	—	●	○	●
Operator semigroup	○	○	—	—	—	—	●
Forward-inv. kernel	●	●	○	—	—	○	●
Multi-agent	—	—	○	●	—	○	●
Nash equilibrium	—	—	—	○	—	○	●
Fixed-point exist.	○	○	○	—	○	○	●
Partial obs.	—	—	○	—	●	○	●
Trajectory safety	●	●	●	○	○	●	●
Operator safety	○	○	—	—	—	—	●
Decidability (PSPACE)	—	—	—	○	○	—	●
Coverage	4/13	5/13	5/13	3/13	6/13	5/13	**13/13**

● fully addressed; ○ partially addressed; — not addressed.

Appendix G Worked Examples

G.1 Example 1: Robotic Manipulation with Safe Operator Learning

Example G.1 (Robotic Arm Governance)**.** Let $X = \mathbb{R}^{12}$ (joint positions, velocities, torques, end-effector pose), $S = \{x \in X : \text{joint torques within limits, no collision}\}$, $Y = \mathbb{R}^6$ (partial proprioceptive observations), $|I| = 1$.

GF instantiation:

- O: Gaussian observation kernel, $o(x, y) \propto \exp(-\|y - Cx\|^2/2\sigma^2)$.
- b_t: Kalman filter posterior (Gaussian belief, Corollary C15).
- $T_t = \pi_\theta(b_t)$: a neural operator (Corollary C19) trained to minimize trajectory cost while satisfying $T_t \in \mathcal{T}_{\text{safe}}$ via a safety layer [?].
- $K^\star$: computed offline via Hamilton-Jacobi reachability [?].

Theorem 4.1 guarantees $x_t \in S$ for all t; Theorem 4.5 guarantees Kalman consistency; Corollary C3 establishes the safe-RL reduction.

G.2 Example 2: Multi-Agent Data Pipeline Governance

Example G.2 (Data Governance as GF)**.** Let X = data schema space, Y = validation reports, $I = \{$producers, consumers, auditors$\}$, $S = \{$schemas satisfying GDPR and quality constraints$\}$.

- T_t = data transformation operator (ETL pipeline).
- $\mathcal{T}_{\text{safe}}$ = operators that preserve schema compatibility and data quality invariants.
- $K^\star$ = database states satisfying all regulatory constraints.
- Equilibrium T^* = consensus schema agreed upon by all agents (Theorem 4.4, Corollary C20).

G.3 Example 3: AI Model Lifecycle Governance

Example G.3 (AI Lifecycle as GF)**.** Let X = model parameter space ($\theta \in \mathbb{R}^d$), b_t = epistemic belief over model performance on deployment distribution, T_t = retraining/fine-tuning operator, S = models satisfying accuracy, fairness, and robustness constraints.

- $\mathcal{T}_{\text{safe}}$ = operators that do not degrade any safety-critical evaluation metric below threshold.
- $K^\star$ = model states jointly satisfying all regulatory and technical requirements.
- Theorem 4.1: retraining never violates a safety constraint if $T_t \in \mathcal{T}_{\text{safe}}$.
- Theorem 4.5: as deployment data accumulates, belief b_t converges to true model performance μ_t (Corollary C6, Theorem 4.5).